Prakash V.S. Tomar

Generative AI Engineer | Agentic Systems | LLMOps & MLOps

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Experience

- **Tata Consultancy Services (TCS)** — Jul 2024 – Present
  Generative AI Engineer — *Gandhinagar, India*
  - **AI Form Builder (Flagship DigiBOLT™):** Architected an **multi-agent AI pipeline** auto-generating enterprise-grade forms from natural language prompts, Excel wireframes, or even HTML — covering document parsing, component selection (**25+ UI components**), validation inference, ANN-based combo population, and schema mapping. Supports create, edit, and incremental add workflows.
  - **LLM Guardrails & Content Safety:** Designed modular safety filters for PII masking, toxicity, bias, and gibberish detection — ensuring **100% data protection** across **10+ enterprise AI** use cases.
  - **LLM Observability & Monitoring:** Instrumented **35+ LLM/ML metrics** via OpenTelemetry, Jaeger, MLflow, LangKit, and Evidently AI — enabling drift detection, distributed tracing, and real-time logging.
  - **Automated Grievance Redressal System:** Production AI pipeline for **3.5M+ users** — two-layer ML classification feeding a **hierarchical RAG system** with **1L+ clustered embeddings** for hallucination-resistant responses in **2–3 seconds**.

  Tech: *FastAPI, LangChain, LangGraph, HuggingFace Transformers, Guardrails, MCP, MLflow, Evidently AI, Jaeger, LangKit, OpenTelemetry, Loguru, Ollama, Azure OpenAI, MongoDB, PostgreSQL, Linux*

- **HappyMonk.ai** — Oct 2022 – Apr 2023
  Data Science Intern (Remote)
  - **Automation Pipelines:** End-to-end pipeline for annotated dataset ingestion, **reducing data preparation time by 45%**.
  - **Model Optimization & Annotation:** Improved object detection and activity recognition by **18%** via YOLOv5 and MMAction2; annotated **6000+** media files.
  - **Geospatial Metadata:** Enhanced pothole density estimation by **30%** via GPS extraction and manipulation.

  Tech: *YOLOv5, MMAction2, DeepFace, PyTorch, PyMongo, MongoDB, MySQL, CVAT, EXIF Tool*

Projects

- **RoadGuard** — Telangana Govt. Smart Pothole Detection — Research / Github
  - Deployed YOLOv8-based road analysis system identifying **3000+ potholes** across Telangana regions.
  - Initiative contributed to a **25% reduction in road accidents** in targeted areas.
- **Technical-Repo Analyzer** — AI Code Complexity Analyzer — Demo / Github
  - Full-stack Python app integrating OpenAI API, GitHub API, and LangChain for automated code analysis via a Streamlit interface.
  - Automated error detection saving **20+ hours per code review cycle**.

Skills

- **Programming Languages:** Python, SQL, Bash, Java
- **Generative AI / Agentic Frameworks:** LangChain, LangGraph, Guardrails, MCP, Ollama, CrewAI, Azure OpenAI Service
- **Observability / MLOps:** MLflow, Evidently AI, Jaeger, OpenTelemetry, LangKit, Loguru
- **Machine Learning / Deep Learning:** Scikit-Learn, Pandas, NumPy, TensorFlow, Keras, PyTorch
- **Backend / Databases:** FastAPI, Streamlit, PostgreSQL, MongoDB
- **Developer Tools:** Linux, Git, VSCode, Jupyter

Education

- **Institute of Engineering and Management** — Kolkata, India
  *B.Tech — Computer Science, Specialization: AI/ML | GPA: **9.26*** — *Nov 2020 – Aug 2024*

  Courses: *Machine Learning, Advanced Deep Learning, AI, DBMS, Data Structures, Algorithms, Data Warehousing*

Honors & Awards

TCS AI Fridays, 2025 — Winner, Gujarat Region; **National Finals (Top 25 teams globally).** Certificate

Amazon ML Challenge, 2023 — Ranked **301st out of 4898 teams** nationally.

Publication

Machine Learning — A New Perspective - Link — Certificate

Presented at IEMPOWER 2022; ML fundamentals and inference vs. predictability tradeoffs. *(In Publication-AIP Publishing)*

Leadership & Community

ML Co-Lead, IEM Coding Club — Mentored **300+ students** in applied DS/ML. *Aug 2022 – Aug 2024*

AI/ML Core Member, GDSC IEM — ML workshops with **210+ participants**. *Aug 2022 – Jul 2023*